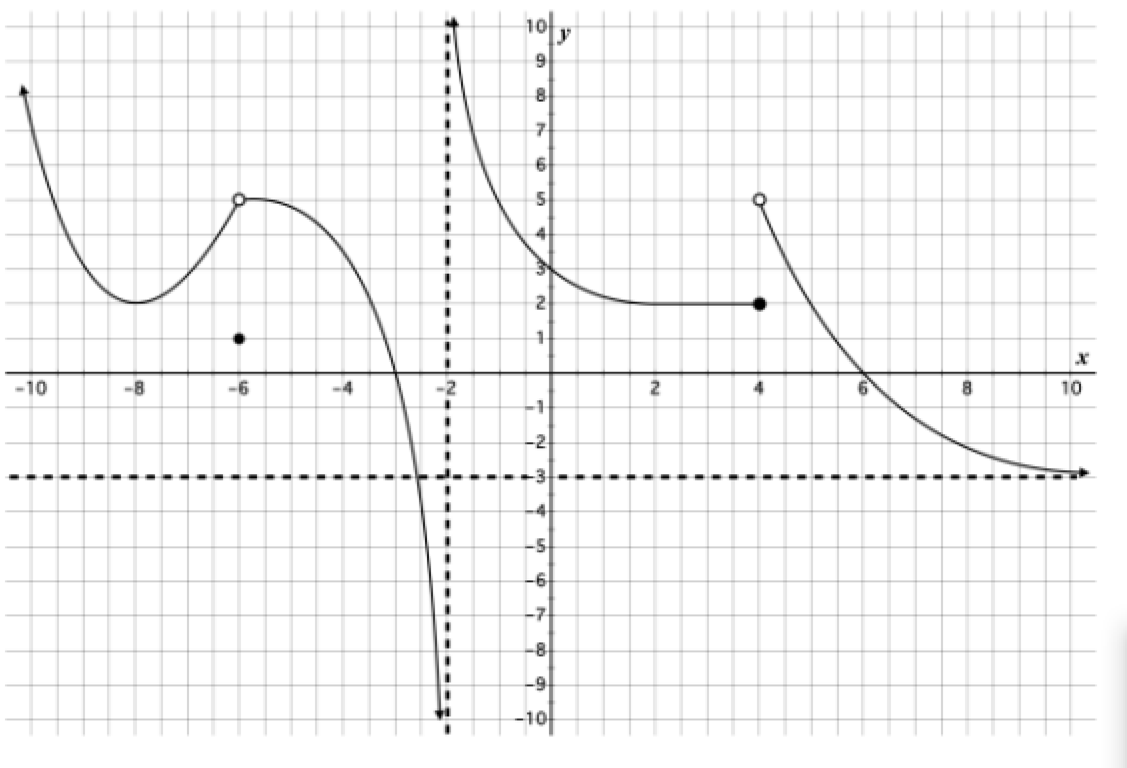
UNIT CODE: BMA 1102 **UNIT NAME: CALCULUS I**

CAT 2 **MON BSCCS GROUP** **28 TH JULY 2025**

INSTRUCTIONS: ANSWER ALL QUESTIONS

1. Find the coordinates of the turning points on the graph of the equation $f(x) = 3x^3 + 8x^2 - 4x + 1$ and distinguish them. (6 Marks)
2. Air is being pumped into a spherical balloon at a rate of 5.6 cubic feet per minute. Find the rate of change of radius when the radius is 4 feet. (4 Marks)
3. An automobile velocity starting from rest is $v(t) = \frac{100t}{2t+16}$ where V is measured in meters per second. Find:
   i. How long does it take for the automobile to reach a velocity of 10m/s (2 Marks)
   ii. Its velocity during the 10th second. (3 Marks)
   iii. Its acceleration after 2 seconds. (5 Marks)
4. If $y = xe^{2x}$ prove that; $\frac{d^2y}{dx^2} + 4y = 4\frac{dy}{dx}$ (5 Marks)
5. Answer the following questions based on the graph below. (5 Marks)



i. $f(8)$

ii. $f(-6)$

iii. $\lim_{x\to 6} f(x)$

iv. $\lim_{x\to 4^-} f(x)$

v. $\lim_{x\to 4^+} f(x)$